W-R-001404B(GSA-FSS)
AMENDMENT-2
December 20, 1973
SUPERSEDING
AMENDMENT
June 4, 1973

AMENDMENT

TO

INTERIM FEDERAL SPECIFICATION

RECORDER-REPRODUCER, SOUND
(PORTABLE, BATTERY/AC POWERED, CASSETTE TYPE)

This amendment was developed by the General Services Administration, Federal Supply Service, Washington, D. C. 20406, based upon currently available technical information. The General Services Administration has authorized this amendment as a part of Interim Federal Specification W-R-001404B (GSA-FSS) dated February 4, 1972.

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Paragraph 1.2.1, Type I: Delete "built-in and remote microphones" and substitute "built-in microphone, remote control microphone".

Paragraph 1.2.1: Add a type VI as follows:

VI: Automatic record level control machine with built-in microphone, remote control microphone, and remote foot start/stop controls.

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Paragraph 2.1: Under Federal Specifications, change "PPP-B-850" to "PPP-C-850".

Paragraph 2.2: Under American National Standards Institute (ANSI) Inc. Standards, add the following publication:

C18.1 - 1972 - Specifications for Dry Cells and Batteries

Paragraph 2.2: Under International Electrotechnical Commission (IEC) Standards, add the following publications:

Amendment No. 1, dated February 1971, to Publication 94 (1968)

94A - 1972 - First Supplement to Publication 94 (1968), Cassette for Commercial Tape Records and Domestic Use, Dimensions and Characteristics.

Paragraph 3.1: Add the following subparagraph:

3.1.1 Requirements for type VI units. All requirements specified herein for a type I unit shall apply equally to a type VI unit except that a backspace review control shall not be required.

FSC 5835

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Paragraph 3.2, line 5: Delete "with reexamination" and substitute "mark".

Paragraph 3.2, line 6: Delete "will" and substitute "may".

Paragraph 3.2.1, line 1: Delete "with reexamination" and substitute "mark".

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Paragraph 3.5: Delete the paragraph in its entirety and substitute the following paragraph:

3.5 Cassette type. A recorder-reproducer furnished under this specification shall be designed to operate with a tape cassette of the co-planar type conforming to IEC Publication 94 (1968) and Amendment No. 1 dated February 1971, and to IEC Publication 94A - 1972, First Supplement to IEC Publication 94 (1968).

Paragraph 3.6.1.1: Change the size requirements for the type I and III units to the following:

Type I - 13 (33.0 centimeters) by 9 (22.9 centimeters) by 4 (10.2 centimeters) inches.

Type III - 10.5 (26.7 centimeters) by 7 (17.8 centimeters) by 3.5 (8.9 centimeters) inches.

Paragraph 3.6.2, line 1: Delete "type III, IV, and V" and substitute "type III and IV".

Paragraph 3.6.2.1, lines 2 and 3: Delete "type III, IV, and V" and substitute "type III and IV".

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Paragraph 3.6.4, 3.6.4.1, and 3.6.4.2: Delete the paragraphs in their entirety and substitute the following paragraphs:

3.6.4 Power supply. Each recorder-reproducer shall be capable of operation in conformance with all requirements of this specification from both battery and AC power sources in accordance with 3.6.4.1 and 3.6.4.2.

3.6.4.1 Batteries. The recorder-reproducer shall utilize either Leclanche (carbon-zinc) or nickel-cadmium rechargeable cells (supplier option) conforming to requirements of ANSI Standard C18.1. The recorder-reproducer shall be designed to contain the batteries in an internal compartment of the basic unit. Batteries need not be supplied with the unit.

3.6.4.2 AC adaptor. The recorder-reproducer shall be provided with an adaptor which enables operation from a 110 - 120 volt, 60 Hz AC power source. The adaptor may be an integral part of the basic unit.

Paragraph 3.6.8.4, lines 2 and 3: Delete the second sentence of the paragraph.

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Paragraph 3.6.10, line 1: After "headphones", delete the remainder of the sentence.

Paragraph 3.6.12: Delete the paragraph in its entirety.

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Paragraph 3.6.16. Delete the paragraph in its entirety and substitute the following paragraph:

3.6.16 Battery check indicator. With the exception of the type V unit, each recorder-reproducer shall be provided with a device which can visually or audibly indicate whether or not the batteries are capable of supplying the power necessary for the unit to perform at a level in accordance with the requirements of this specification.

Paragraph 3.7: Delete the paragraph in its entirety and substitute the following paragraph:

3.7 Performance requirements. Unless otherwise specified, the requirements of this section shall apply to the performance of each type of recorder-reproducer when operating with the required method(s) of record level control. Test input level, as referred to in this section, is defined in 4.6.1.4 and 4.6.1.5.

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Paragraph 3.7.4, 3.7.5, 3.7.6, 3.7.7, and 3.7.8: Delete the paragraphs in their entirety and substitute the following paragraphs:

3.7.4 Reproduce amplifier output (rated output). When a 333 Hz signal input to the microphone input jack (and line input jack if such jack is required by this specification) has been recorded at test input level, the reproduce amplifier shall be capable of producing a minimum required power output to a purely resistive load equal to the manufacturer's stated output impedance. The minimum required power output shall be 200 milliwatts for the type V unit and shall be 300 milliwatts for all other units. As used herein, the term "rated output" shall mean the output voltage which produces the minimum required power output to a purely resistive load equal to the manufacturer's stated output impedance.

3.7.5 Harmonic distortion. When a 333 Hz sine wave signal input to the microphone input jack (and line input jack if such jack is required by this specification) is recorded at test input level and reproduced to obtain a reproduce amplifier output equal to rated output, the overall record-reproduce harmonic distortion shall be no more than 6 percent for a type V unit and no more than 5 percent for all other units when measured as specified in 4.8.6.1. When a 333 Hz sine wave signal input to the line input jack, if such jack is required by this specification, has been recorded at an input level of 1 volt and reproduced to obtain a reproduce amplifier output 6 dB below rated output, the overall record-reproduce harmonic distortion shall be no more than 5 percent when measured as specified in 4.6.8.2.

3.7.6 Reproduce system frequency response. When compared to the response at 1000 Hz, the reproduce system frequency response in playback of the level control test tape (see 4.6.1.1) shall be within ± 5 dB over the frequency range 125 to 6300 Hz when tested as specified in 4.6.9.

3.7.7 Overall frequency response. When compared to the response at 1000 Hz, the overall record-reproduce frequency response, including tape, shall be within ± 5 dB over the range 125 to 6300 Hz when tested as specified in 4.6.10.

3.7.8 Signal-to-noise ratio. The ratio of signal level to unweighted noise level shall be at least 40 dB when measured as specified in 4.6.11.

Paragraph 3.7.9, line 2: Delete "35 dB" and substitute "50 dB".

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Paragraph 4.6.1.1.2, item a: Delete item "a" in its entirety and substitute the following:

a. Recorded tape flux characteristic (record equalization) as specified by IEC Publication 94 (1968), and Amendment No. 1 dated February 1971, and IEC Publication 94A - 1972, First Supplement to IEC Publication 94 (1968).

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Paragraph 4.6.1.2: Delete the second sentence of the paragraph and substitute the following sentence:

All output voltage levels shall be measured across a purely resistive load equal to the manufacturer's stated output impedance and connected to the reproduce amplifier output with the speaker disconnected.

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Paragraph 4.6.9, line 2: Insert "500", between "250," and "1000".

Paragraph 4.6.9, lines 2 and 3: Delete the phrase "and for a type II unit, the 8000 Hz signal".

Paragraph 4.6.10, line 3: Delete the phrase "and for a type II unit, the 8000 Hz signal".

Paragraph 4.6.10, line 6: After the end of the third sentence, insert the following provision:

"(An alternate means of eliminating signal input to the microphone jack shall be used when removal of the microphone plug from the jack would automatically connect a built-in microphone to the input)".

Paragraph 4.6.12: Delete the paragraph in its entirety and substitute the following paragraph:

4.6.12 Erasure. Record a blank test tape with a 333 Hz signal at a level of input to the microphone input jack which will produce 3 percent third harmonic distortion when reproduced at an output level 20 dB below rated output. Play the recorded signal and adjust the reproduce volume control to obtain an output 3 dB below rated output when measured with a tuneable voltmeter tuned to 333 Hz. Erase the recorded signal and reproduce the erased portion of the tape noting the reduction in output level as measured by the tuneable voltmeter to determine compliance with 3.7.9. (It is convenient to perform this test with the recording used in 4.6.11.)

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